

VisaPath — Project Status & Notes

Last Updated: February 16, 2026 **Hackathon:** DevDash 2026 (Deadline: Feb 20) **Live URL:** <https://visapath-app.azurewebsites.net> **GitHub:** <https://github.com/Dileep2896/visapath>

What's Done

Core Features

- **AI Timeline Generation** — Gemini 2.5 Flash (fallback to 2.0 Flash) generates personalized immigration timelines from 15+ profile fields
- **RAG-Augmented Chat** — 8 knowledge-base docs embedded in ChromaDB, top-k=4 retrieval, sourced responses
- **Risk Analysis Engine** — CPT overuse, country backlog, OPT deadline, unemployment limits, H-1B uncertainty, cap-gap expiration
- **What-If Simulator** — Modify profile inputs and re-run timeline without consuming a credit
- **Tax Guide** — AI-generated tax filing guide with residency/FICA determination
- **Document Tracker** — Checklists for OPT, STEM OPT, H-1B, Green Card steps
- **Action Items** — Urgency-sorted checklist derived from timeline events

Auth & Admin

- Email/password registration with bcrypt hashing + JWT auth
- Per-IP rate limiting on auth endpoints (10 req/min)
- Per-user AI credit system (default limit: 5, admin-overridable)
- Admin dashboard: user CRUD, credit management, limit overrides
- Demo account seeded: `demo@visapath.com` / `demo123`
- Admin account seeded via env vars

Frontend

- React 19 + TypeScript + Vite + Tailwind CSS v4
- Dark theme with teal accents, smooth animations
- 4-step onboarding form with draft save/resume
- Vertical timeline with expandable cards, staggered animations, urgency color-coding
- Mobile-responsive layout with hamburger menu
- Skeleton loaders, toast notifications, empty states
- Admin skips onboarding — goes straight to admin dashboard

Backend

- Python FastAPI with Gunicorn + Uvicorn (2 workers)
- Dual database: SQLite (dev) / PostgreSQL (prod via DATABASE_URL)
- Schema auto-migration on startup
- pysqlite3 patch for Azure's old SQLite (ChromaDB compat)
- Security headers: nosniff, DENY framing, HSTS, strict referrer

Deployment

- Azure App Service (Free F1, East US, Python 3.11)
 - Monolith: FastAPI serves built React SPA from `backend/static/`
 - GitHub Actions CI/CD on push to `main`
 - Environment vars configured in Azure App Settings
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What's NOT Done / Improvements

High Priority (before submission)

- ☐ Record demo video (3 min — see plan in `VisaPath_Build_Plan.md`)
- ☐ Write Devpost submission text
- ☐ Final testing pass across all features
- ☐ Mobile responsiveness polish pass

Known Issues

- ☐ Document tracker checkbox state is NOT persisted (client-side only, resets on refresh)
- ☐ Rate limiting is in-memory only — resets on server restart, not shared between Gunicorn workers
- ☐ Legacy `timeline_generator.py` (rule-based) still exists alongside `ai_timeline_generator.py` — can be deleted
- ☐ `eslint-disable` type escape in TaxGuidePage callback typing

Missing Features (from build plan)

- ☐ Saved timeline history UI — backend endpoint (`GET /api/auth/my-timelines`) exists but no frontend page
- ☐ Calendar export — export deadlines to Google Calendar (.ics)
- ☐ Email/push reminders for upcoming deadlines
- ☐ PDF export of timeline and checklist
- ☐ Employer E-Verify lookup
- ☐ Community mode (anonymized stats)

Partial Implementations

- ☐ H-4 EAD — selectable in form but no dedicated timeline logic
- ☐ L-1 visa — selectable but no specific rules/risk analysis
- ☐ J-1 academic training — selectable but minimal pathway coverage
- ☐ Tax treaty coverage — relies on Gemini knowledge, no hardcoded treaty database
- ☐ "Other" countries — only India, China, Rest of World have specific backlog data

Nice-to-Have Improvements

- ☐ Persist document tracker checkbox state to DB
- ☐ Shared rate limiting (Redis or DB-backed) across workers
- ☐ Stripe/payment integration for premium credits
- ☐ Dark/light mode toggle
- ☐ Accessibility audit (ARIA labels, keyboard navigation)
- ☐ E2E tests (Playwright/Cypress)
- ☐ API integration tests

Key Files Reference

Purpose	Path
FastAPI entry point	backend/app/main.py
Database schema + CRUD	backend/app/database.py
AI timeline generator	backend/app/services/ai_timeline_generator.py
Gemini wrapper (new google.genai)	backend/app/services/gemini_service.py
Risk analyzer	backend/app/services/risk_analyzer.py
RAG pipeline	backend/app/services/rag_service.py
Auth service	backend/app/services/auth_service.py
Immigration rules data	backend/app/data/immigration_rules.py
RAG knowledge base (8 docs)	backend/app/rag/documents/
React app + routes	frontend/src/App.tsx
Auth context (global state)	frontend/src/context/AuthContext.tsx
TypeScript types	frontend/src/types/index.ts

Purpose	Path
API client	frontend/src/utils/api.ts
CI/CD pipeline	.github/workflows/deploy.yml
Azure config	backend/.azure/config
Startup command	backend/startup.sh

Environment Variables (Azure App Settings)

Variable	Purpose
GEMINI_API_KEY	Google Gemini API key
JWT_SECRET	JWT signing secret
DATABASE_URL	PostgreSQL connection string (omit for SQLite)
CORS_ORIGINS	Comma-separated allowed origins
ADMIN_EMAIL	Admin account email (seeded on startup)
ADMIN_PASSWORD	Admin account password (seeded on startup)

Recent Changes (Feb 16)

- Migrated from deprecated `google.generativeai` to new `google.genai` SDK
- Removed `google-generativeai` from requirements.txt
- Admin now skips onboarding — redirected to `/admin` on login and on direct navigation